

REST API & JSON DATA EXTRACTION

USD exchange rates converted from JSON to analysis-ready CSV

Bluestock FinTech Internship | Week 2 Task 2 | 12 September 2026

Assignment result

A public REST API was called with the HTTP GET method. The JSON response was inspected, validated and converted into a clean CSV dataset for analysis. The extraction covers daily USD exchange rates against INR, EUR and GBP from 12 August to 11 September 2026.

HTTP RESPONSE 200 OK Request completed	VALIDATED RECORDS 93 31 dates x 3 currencies	MISSING VALUES 0 All required fields present
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1. API selected

Item	Value
API	Frankfurter exchange-rate API, version 2
Base endpoint	https://api.frankfurter.dev/v2/rates
HTTP method	GET - used to retrieve data without changing the server
Authentication	None - no API key or login required
Tools	Postman collection for the request and tests; Python for JSON-to-CSV conversion
Response format	JSON array containing date, base, quote and rate
Data scope	USD to INR, EUR and GBP; 2026-08-12 to 2026-09-11

2. Request and query parameters

```
GET https://api.frankfurter.dev/v2/rates
?from=2026-08-12
&to=2026-09-11
&base=USD
&quotes=INR,EUR,GBP
```

- from and to define the historical period.
- base=USD means every rate starts with one US dollar.
- quotes limits the response to INR, EUR and GBP, keeping the dataset focused.
- The server returned HTTP status 200, which means the request succeeded.

3. Inspecting the JSON response

JSON stores data as structured key-value pairs. In this response, the top-level value is an array. Each object inside the array represents one currency rate on one date.

```
[
{
  "date": "2026-08-12",
  "base": "USD",
  "quote": "EUR",
  "rate": 0.86645
},
...
]
```

JSON field interpretation

JSON field	Meaning	Example	Data type
date	Date of the exchange rate	2026-08-12	String
base	Currency being converted	USD	String
quote	Currency received	INR	String
rate	Units of quote currency for 1 base unit	95.37	Number

4. Converting JSON into CSV

The Python script reads the JSON array, checks every record and creates a rectangular table. The field names are made clearer for analysis: base becomes base_currency, quote becomes quote_currency, and rate becomes exchange_rate.

date	base_currency	quote_currency	exchange_rate
2026-08-12	USD	EUR	0.86645
2026-08-12	USD	GBP	0.73964
2026-08-12	USD	INR	95.37

Validation performed

Check	Result
Top-level response is an array	Passed
Required fields exist in every object	Passed
Base currency is always USD	Passed
Quote currency is INR, EUR or GBP	Passed
Exchange rates are numeric and greater than zero	Passed
Rows sorted by date and currency	Passed
Final CSV record count	93

5. Analysis of the CSV dataset

The table compares the first and last observed rate and summarizes the 31 daily values for each quote currency. The percentage change is calculated as: $(\text{last rate} / \text{first rate} - 1) \times 100$.

Currency	First	Last	Average	Minimum	Maximum	Change
EUR	0.86645	0.86060	0.86048	0.85553	0.86654	-0.68%
GBP	0.73964	0.73878	0.73724	0.73302	0.74063	-0.12%
INR	95.37	95.46	95.26	94.45	95.70	+0.09%

Beginner interpretation

- The USD bought slightly fewer euros at the end of the period: the EUR rate decreased by 0.68%.
- The GBP rate changed very little, decreasing by 0.12%.
- The USD bought slightly more Indian rupees at the end: the INR rate increased by 0.09%.
- All three changes were small. A 31-day sample is useful for practice but is not enough to predict future currency movements.
- The rate direction must be read carefully: because USD is the base, a higher INR rate means one USD purchases more INR.

6. GET, POST and authentication

Concept	Meaning	How it applies here
GET	Retrieves data from a server	Used to obtain exchange rates
POST	Sends data to create or start a server-side action	Not needed because no new record was created
Authentication	Proves that a client is authorized	Not required by this public endpoint
Query parameter	A key-value filter added after ? in a URL	Used for dates and currencies

7. Deliverables and reproducibility

- Postman Collection: imports the GET request, variables and response tests into Postman.
- Raw JSON: preserves the exact structured response received from the API.
- Clean CSV: can be opened directly in Excel or Google Sheets for analysis.
- Python script: repeats the GET request, validates the response and recreates both data files.
- This report: documents the method, validation, results and limitations.

Conclusion

The assignment demonstrates a complete small API data pipeline: request, JSON inspection, validation, transformation, CSV export and analysis. The outputs are traceable and reproducible, and no confidential credentials were used.

Source: Frankfurter official documentation - <https://frankfurter.dev/>. Data endpoint - <https://api.frankfurter.dev/v2/rates>. Educational analysis only; exchange rates are not financial advice.